



STA201 Assignment 1 Solution (Spring 2023)

Question 1

The following table shows some information on a variety of different vehicles. Using the information given in table 1, answer question 1a – 1d.

Table-1: Cars

Model	Number of gears	Size of Engine	Transmission	Vehicle Class	Highway Mileage	City Mileage	Model Year
Allion	6	3.4	Manual	Midsize Car	22	17	1997
Axio	7	2.5	Manual	Small Car	26	22	2000
Aqua Hybrid	8	3.5	Automatic	Small Car	30	24	2021
Corolla	5	3	Manual	Midsize Car	25	22	2001
Vezel	8	6.8	Automatic	Large Car	18	15	2022
Premio	6	3.3	Automatic	Midsize Car	22	17	2016
Harrier	7	4.2	Automatic	Large Car	18	15	2020
Naoh	7	5.4	Automatic	Large Car	20	16	2018

Question 1(A)

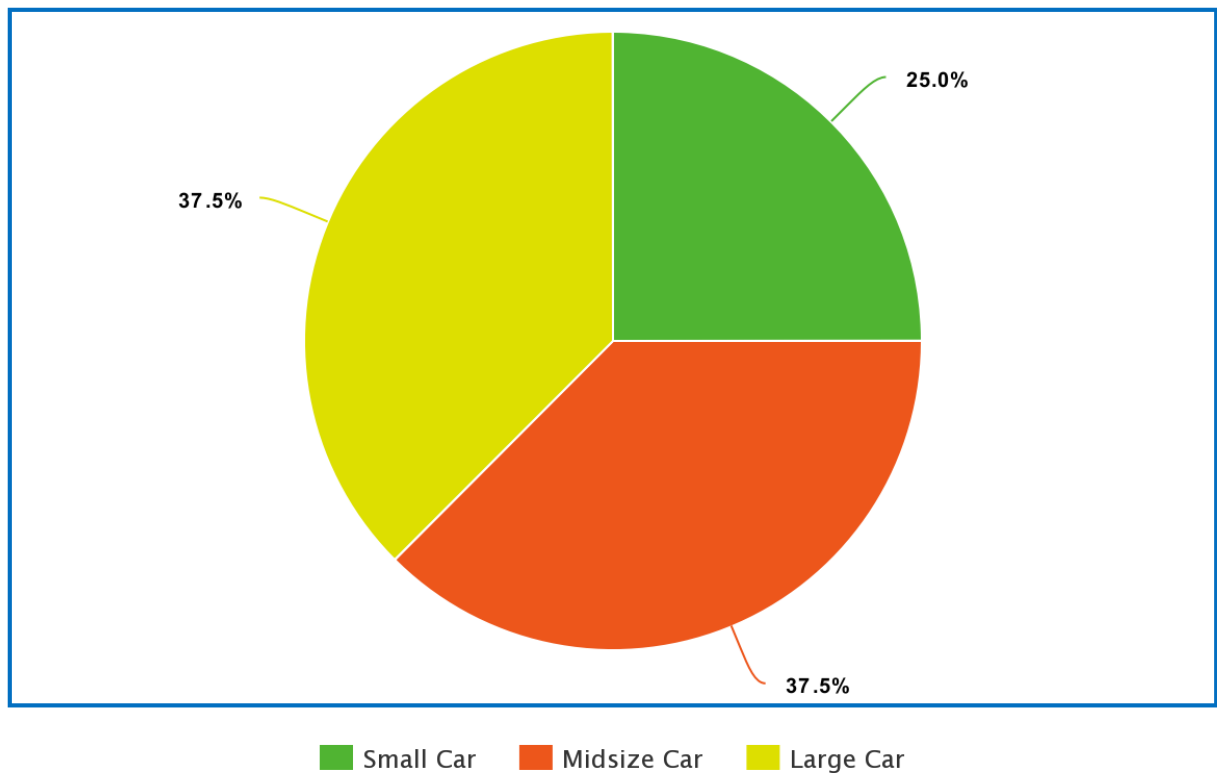
- (i) How many variables are listed in table 1? (**Answer: 8**)
- (ii) Classify the variables according to their types (Qualitative / Quantitative).

Qualitative	Quantitative
Model	Number of gears
Transmission	Size of Engine
Vehicle Class	Highway Mileage
Model Year	City Mileage

- (iii) Construct a frequency distribution table to represent the summary information of the variable “Vehicle Class” and display the results in a pie chart.

Answer:

Vehicle Class	Frequency	Relative Frequency	Angles in degree
Small Car	2	$2/8=0.25$	90
Midsize Car	3	$3/8=0.375$	135
Large Car	3	$3/8=0.375$	135
	Total = 8	Total = 1	Total = 360

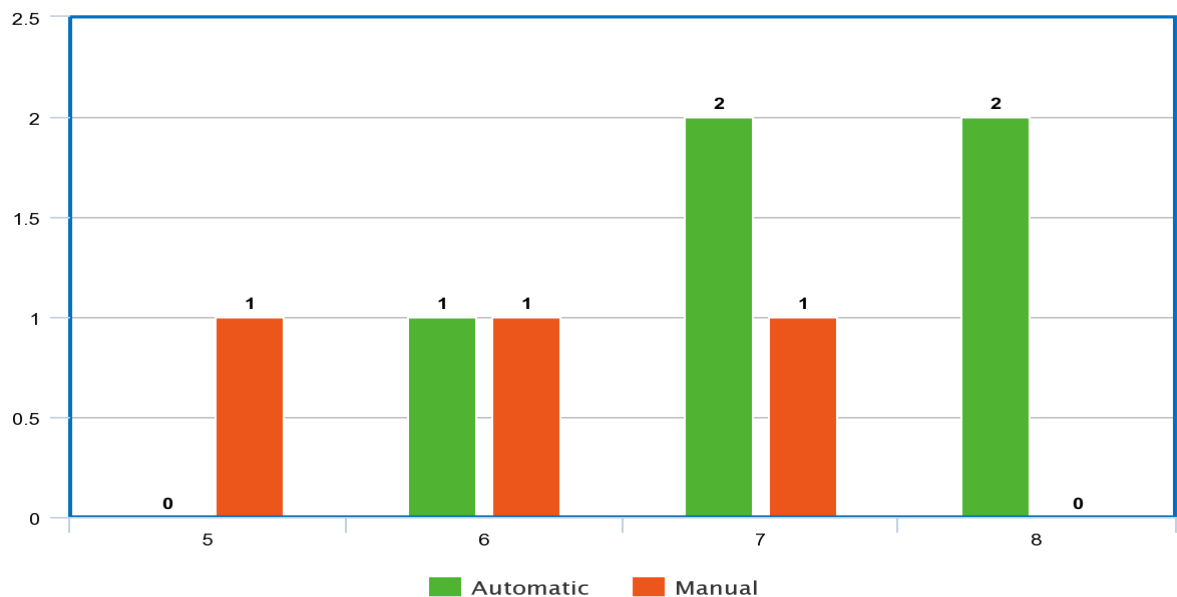


Question 1(B)

Complete the following table and answer the questions:

Transmission	Number of Gears				Total
	5	6	7	8	
Automatic	0	1	2	2	5
Manual	1	1	1	0	3
Total	1	2	3	2	8

- (i) What is the modal response for the variable “Transmission”? (Which has the highest frequency?) (Answer: Automatic)
- (ii) What proportion of vehicles have seven gears? (Answer: $3/8$ or 37.5%)
- (iii) What proportion of Automatic vehicles have eight gears? (Answer: $2/5$ or 40%)
- (iv) What proportion of vehicles with six gears are Manual? (Answer: $1/2$ or 50%)
- (v) Construct a side-by-side bar chart to represent the information given in table 2.



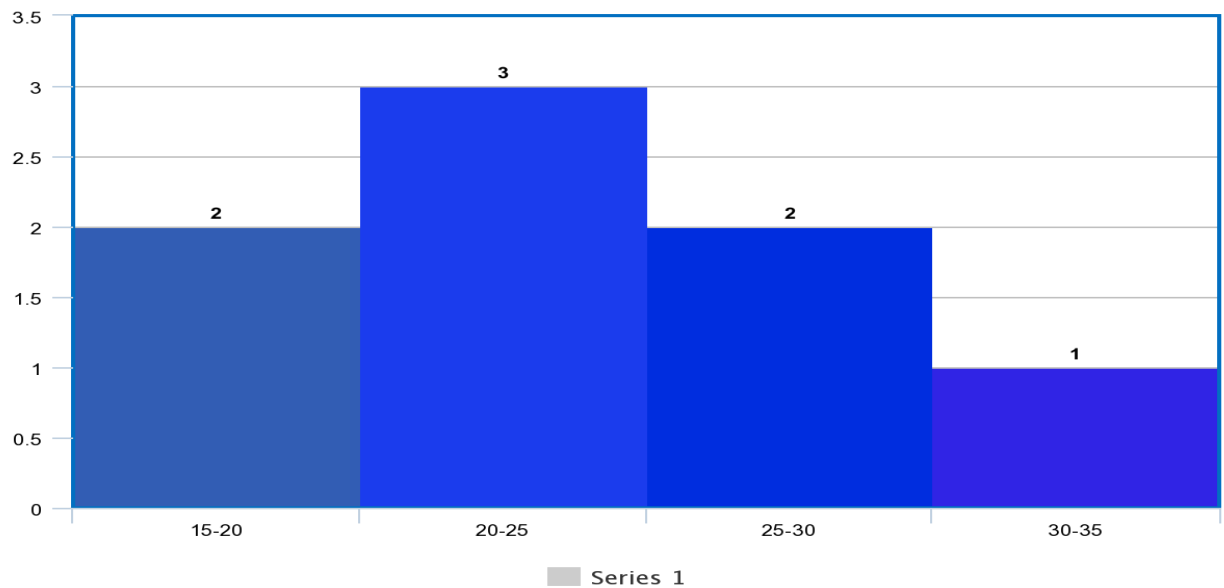
Question 1(C)

Complete the following table and answer the questions:

Table 3: Frequency distribution of Highway Mileage

Highway Mileage	Tally	Frequency	Relative frequency	Cumulative relative frequency
15 – 20		2	$2/8 = 0.25$	$2/8$
20 – 25		3	$3/8 = 0.375$	$5/8$
25 – 30		2	$2/8 = 0.25$	$7/8$
30 – 35		1	$1/8 = 0.125$	$8/8$

- (i) What proportion of vehicles have mileage between 20 and 30 Highway Mileage?
(Answer: $5/8$ or 62.5%)
- (ii) What proportion of vehicles have mileage of at least 30 Highway Mileage?
(Answer: $1/8$ or 12.5%)
- (iii) Construct a histogram to display the data represented in table 3.



Question 2

Recall that 65 purchasers have participated in a survey and have rated the XYZ-Box video game system. The composite ratings that have been obtained are as follows:

39	38	40	40	40	46	43	38	44	44	44
45	42	42	47	46	45	41	43	46	44	42
38	46	45	44	41	45	40	36	48	44	47
42	44	44	43	43	46	43	44	44	46	43
42	40	42	45	39	43	44	44	41	39	45
41	39	46	45	43	47	41	45	45	41	

(a) Construct a stem-and-leaf display for the 65 composite ratings.

Answer:

Stem	Leaf
3	9,8,8,8,6,9,9,9
4	0,0,0,6,3,4,4,4,5,2,2,7,6,5,1,3,6,4,2,6,5,4,1,5,0,8,4,7 , 2,4,4,3,3,6,3,4,4,6,3, 2,0,2,5,3,4,4,1,5,1,6,5,3,7,1,5,5,1

(b) If we consider a purchaser to be “very satisfied” if his or her composite score is at least 42, can we say that almost all purchasers of the XYZ-Box video game system are “very satisfied”. Describe in your own words.

Answer: Here, we can consider a purchaser very satisfied if his/her composite score is ≥ 42 . However, if we look into the XYZ-Box video game system, we can see that not all the ratings are greater than or equal to 42. Some ratings are below 42. Such as, 36,38,39, 40, 41 are the ratings below 42. Around 19 ratings are below 42. To say almost all the purchasers are very satisfied, then all the ratings should be greater than or equal to 42. In that case, as not all the ratings are ≥ 42 , we cannot say that almost all purchasers of the XYZ-Box video game system are “very satisfied”.

Question 3

You are given the following table where the number of stocks falling in the particular price categories in the Chicago Stock Exchange (CSE):

Stock Prices (\$)	No. of Stocks
50 – 60	8
60 – 70	$\sqrt{144}$
70 – 80	19
80 – 90	24
90 – 100	$\sqrt{25} + \text{X} + \text{Y}$

Here, **X** & **Y** are the **last two digits** of your student ID (i.e., 191000**12**, 201234**78**, etc.). [If your ID is 191000**12** then, the No. of Stocks in the last row (90 – 100) will be $\sqrt{25} + \text{1} + \text{2} = 8$]. Estimate the following:

a) Find:

- i. Mean
- ii. Median
- iii. Mode

a) For **X** = 9, **Y** = 9

Frequency of class (90 – 100) = 5 + 9 + 9 = 23

Complete frequency distribution table:

	Stock Prices (\$)	Frequency (f_i)	Mid value (x_i)	Cumulative frequency	$f_i x_i$
	50 - 60	8	55	8	440
	60 - 70	12	65	20	780
	70 - 80	19	75	39	1425
Median & Modal Class	80 - 90	24	85	63	2040
	90 - 100	23	95	86	2185
		$\sum_{i=1}^5 f_i = 86$			$\sum_{i=1}^5 f_i x_i = 6870$

i) Mean:

$$\underline{\underline{x}} = \frac{\sum_{i=1}^5 f_i x_i}{\sum_{i=1}^5 f_i} = \frac{6870}{86} = \mathbf{79.88}$$

The mean is **79.88**

ii) Median:

$$\frac{n}{2} = \frac{86}{2} = 43$$

$$L_m = 80$$

$$F = 39$$

$$f_m = 24$$

$$c = 10$$

$$\text{Median} = L_m + \frac{\frac{n}{2} - F}{f_m} * c = 80 + \frac{43 - 39}{24} * 10 = \mathbf{81.67}$$

The median is **81.67**

iii) Mode:

$$L_0 = 80$$

$$\Delta_1 = 24 - 19 = 5$$

$$\Delta_2 = 24 - 23 = 1$$

$$c = 10$$

$$\text{Mode} = L_0 + \frac{\Delta_1}{\Delta_1 + \Delta_2} * c = 80 + \frac{5}{5 + 1} * 10 = \mathbf{88.33}$$

The mode is **88.33**

XY	Mean	Median	Mode
00	75.88	77.37	82.08
01/10	76.16	77.63	82.17
02/20	76.43	77.89	82.27
03/30	76.69	78.16	82.38
04/40	76.94	78.42	82.50
05/50	77.19	78.68	82.63
06/60	77.43	78.95	82.78
07/70	77.67	79.21	82.94
08/80	77.89	79.47	83.13
09/90	78.12	79.74	83.33
11	76.43	77.89	82.27
12/21	76.69	78.16	82.38
13/31	76.94	78.42	82.50
14/41	77.19	78.68	82.63
15/51	77.43	78.95	82.78
16/61	77.67	79.21	82.94
17/71	77.89	79.47	83.13
18/81	78.12	79.74	83.33
19/91	78.33	80.00	83.57
22	76.94	78.42	82.50
23/32	77.19	78.68	82.63
24/42	77.43	78.95	82.78
25/52	77.67	79.21	82.94
26/62	77.89	79.47	83.13
27/72	78.12	79.74	83.33
28/82	78.33	80.00	83.57

29/92	78.54	80.21	83.85
33	77.43	78.95	82.78
34/43	77.67	79.21	82.94
35/53	77.89	79.47	83.13
36/63	78.12	79.74	83.33
37/73	78.33	80.00	83.57
38/83	78.54	80.21	83.85
39/93	78.75	80.42	84.17
44	77.89	79.47	83.13
45/54	78.12	79.74	83.33
46/64	78.33	80.00	83.57
47/74	78.54	80.21	83.85
48/84	78.75	80.42	84.17
49/94	78.95	80.63	84.55
55	78.33	80.00	83.57
56/65	78.54	80.21	83.85
57/75	78.75	80.42	84.17
58/85	78.95	80.63	84.55
59/95	79.15	80.83	85.00
66	78.75	80.42	84.17
67/76	78.95	80.63	84.55
68/86	79.15	80.83	85.00
69/96	79.34	81.04	85.56
77	79.15	80.83	85.00
78/87	79.34	81.04	85.56
79/97	79.52	81.25	86.25
88	79.52	81.25	86.25

89/98	79.71	81.46	87.14
99	79.88	81.67	88.33

Question 4

A group of colleagues is going to Safari Park for a day. They arranged their trip into three halves and planned to travel at a speed of 68 kilometers per hour on their journey to their goal. Their top speeds were 72 km/h and 88 km/h for the first and second parts, respectively. If they are to meet their intended average speed, what speed should they maintain throughout the third part of their journey?

Answer:

Using Harmonic Mean:

$$HM = n/(1/s_1 + 1/s_2 + 1/s_3)$$

$$\therefore 68 = 3/\{1/72 + 1/88 + 1/s_3\}$$

$$\therefore s_3 = 53.0079 \text{ km/hr}$$

They should maintain a speed of 53.0079 km/h for the third part of their journey

Question 5

Suppose, you spent Tk. 3,500,000 to buy a new car. After the first year, the car depreciates by 40%, 20% after the second year, and 10% after the third year.

(a) What is the average rate of depreciation per year after three years?

Answer:

Using Geometric Mean:

$$\text{Average rate of depreciation per year after 3 years: } (0.6 \cdot 0.8 \cdot 0.9)^{(1/3)} = 0.7559$$

(b) What will be the value of the car after three years of use?

Answer:

$$\text{Value of car after 3 years: } 35,00,000 \cdot (0.7559)^3 = 35,00,000 \cdot 0.6 \cdot 0.8 \cdot 0.9 = \text{Tk } 1512000$$